



Aung Min Khant

Mechatronics Engineer

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate with hands-on experience in process improvement, system integration, testing, troubleshooting, and technical documentation, supported by professional experience on an 80MW hyperscale data center commissioning project. Skilled in engineering analysis, CAD design, embedded systems, and cross-functional collaboration, with a strong interest in NPI, R&D, manufacturing process optimization, and continuous improvement.

PROFESSIONAL EXPERIENCE

Data Center Commissioning Engineer | Longmotive Construction (Thailand) Co., Ltd. | Nov 2025 – Apr 2026

- Inspection Management (**Malaysia & Thailand**): Conducted delivery inspections of cable trays, VRV AC units, EMSB, RMU, and prefab MVP modules — verifying serial numbers, unit counts, ratings, FAT, and FWT records; verified underground pipe installation alignment against engineering drawings; supported Lock Out Tag Out procedures on-site.
- Commissioning Scripts, MOS & Tracking: Authored, verified, and submitted Cx Level 1 and 2 mechanical commissioning and testing scripts and Method of Statement to clients via **Autodesk Construction Cloud**
- Engineered a custom submittal tracker that eliminated documentation gaps and maintained documentation completeness with zero missing records.
- Digital Integration & Coordination: Optimized project data workflows by integrating Color-Level Tag data into CxPlanner and coordinating with **NV5** consultants to resolve revisions, improve data consistency, and support accurate technical documentation.
- Intern Mentorship & Operational Scale: Scaled field operations by training and leading an intern team, driving the successful execution of 1,000+ delivery inspections with absolute data integrity and zero documentation errors.

ACADEMIC PROJECTS (Note: All project titles are hyperlinked—click to view details on my official personal website.)

Pneumatic Shirt Folding System | Published

- Designed and developed an electro-pneumatic prototype for automated garment folding.
- Integrated mechanical, pneumatic, and control systems and performed functional testing.
- Improved folding reliability and cycle performance through iterative troubleshooting and mechanical refinement.

AI Indoor Localization System

- Developed a 7-node indoor localization system using multi-sensor fusion (LoRa, RSSI, compass, path constraints) and Google Colab AI processing, enabling precise zone-level identification across 4 transmitters.
- Improved positioning resolution from ambiguity between 2–3 transmitter zones to accurate single-node identification by combining unprocessed RSSI values, heading data, and path constraint logic.
- Collected and analyzed sensor data using ESP32 and cloud-based monitoring to support performance evaluation and decision-making.

Advanced Kinetic Energy Harvesting System

- Developed a kinetic energy harvesting system converting industrial vibrations and pedestrian footfall into electrical energy, achieving approximately 4.6W average output (0.675A, 6.875V).
- Conducted testing, data collection, performance analysis, and design improvement to evaluate system reliability and identify causes of performance limitations to capture footfalls at intervals of 1 second, ensuring continuous energy capture from human foot traffic.

Automated Storage and Retrieval System

- Engineered a 3-axis automated storage and retrieval system (ASRS) with a closed-loop motion control workflows to execute precise horizontal alignment, automated Z-axis (prismatic forklift end-effector) cargo insertion/retrieval sequences, and real-time inventory tracking with high spatial repeatability.

Menki Noodle Vending Machine | Ladder Logic and HMI Design

- Engineered a fully simulated autonomous noodle vending machine control system featuring structured Ladder Logic (Omron Cx Programmer) and HMI design (Omron Cx Designer) interfaces to execute sequential workflows.

Room Renovation and 5 Year Design Review

- Executed full-scale interior renovation using real-life dimensional measurements and 3D modeling to optimize layout and storage, followed by a 5-year design audit to evaluate long-term spatial planning, and technical decisions.

GPS Linear Interpolation

- Collected raw location coordinates via a GPS module and ESP32 logged to Google spreadsheets for error analysis, then applied spherical linear interpolation in Matlab to smooth noisy trajectories and restore route accuracy.

TECHNICAL SKILLS

- Automation & Control:** PLC Programming, Robotics, HMI Design, PID Control, Automation Integration
- Embedded Systems & IoT:** ESP32, Arduino, IoT Monitoring, Sensor Integration, LoRa, Motor driving system
- Computer Vision & AI:** OpenCV, Multi-Sensor Fusion, Google Colab AI, Path Constraints
- Electronics & Hardware:** PCBA Testing, Circuit Troubleshooting, Power Electronics, Electrical Schematic Reading
- Design & Structural Analysis:** AutoCAD, SolidWorks, 3ds Max, Stress & Vibration Testing, Mechanical Assembly
- RF & Data Collection:** RF Communication, Wireless Protocols (LoRa), Data Acquisition (DAQ)
- Software & Logic:** Python, C++, Microsoft Visual Studio, MATLAB,
- Data Center Commissioning:** Cx Procedures (Level 1–3), Script Execution, Site Status Tracking, Technical Reporting
- Design & Manufacturing:** GD&T (basic), Prototype Development, Continuous Improvement, Design Validation
- Quality Management:** Root Cause Analysis, End-to-End Validation, Process Improvement, Failure Analysis

TOOLS & SOFTWARE

AutoCAD	SolidWorks	3ds Max	CxDesigner	CxProgrammer	CxPlanner
Python	C++	OpenCV	Visual Studio	MATLAB	Processing
Arduino IDE	Microsoft Excel	Microsoft Word	Microsoft PPT	Adobe Acrobat	Autodesk

EDUCATION

Bachelor of Mechatronics Engineering in Automation | Assumption University, Thailand | 2025 | GPA: 3.12 / 4.00

ACHIEVEMENT (Note: The description is hyperlinked—click it to explore further.)

1st Place – 3D Computer Aided Design of Innovative Car Competition, Assumption University (2025)

LANGUAGES

Myanmar: Native English: Professional Proficiency



Scan the QR code to learn more about my portfolio, background, and work.